Exact values

1 Write down the exact value of the following:

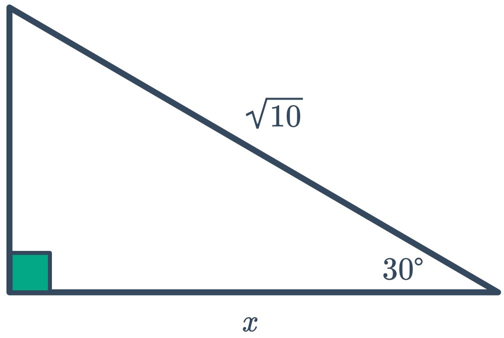
a $\sin 60^\circ$ b $\cos 60^\circ$ c $\tan 30^\circ$ d $\sin 45^\circ$

2 Given that $\sin\theta = \dfrac{1}{2}$ in a right triangle:

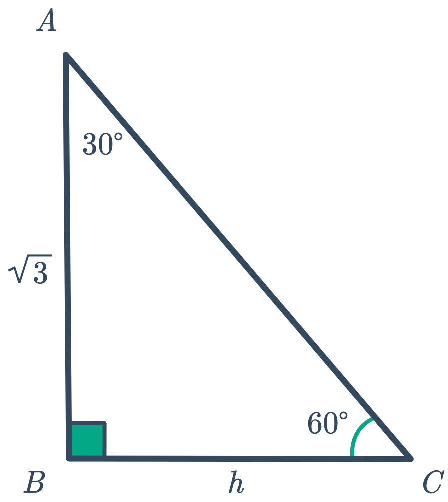
a State the value of θ. b Hence, find $\cos\theta$.

3 Find the exact value of the pronumeral(s) in the following triangles:

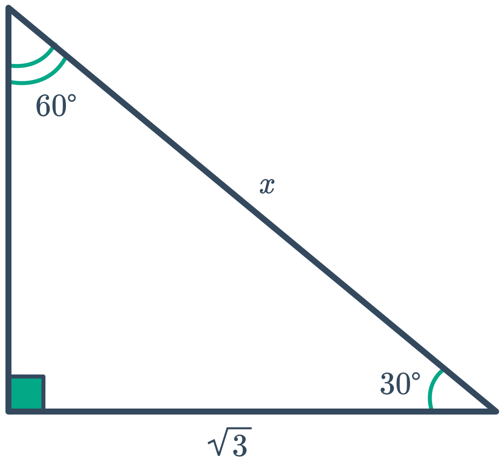
a



b



c



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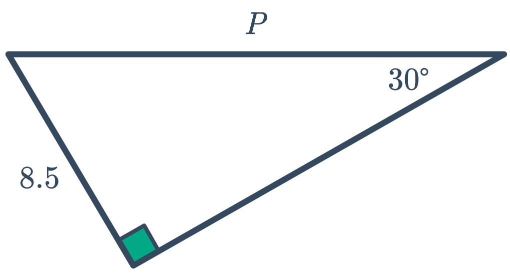
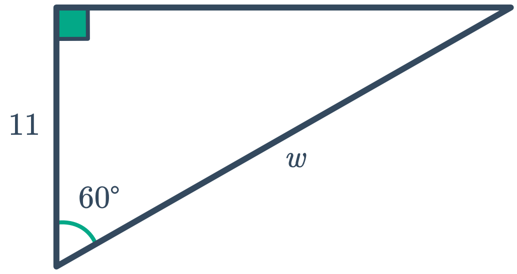
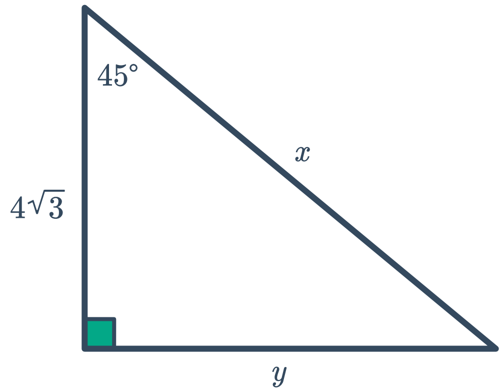


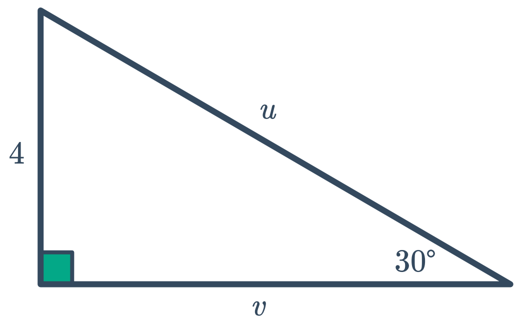
e



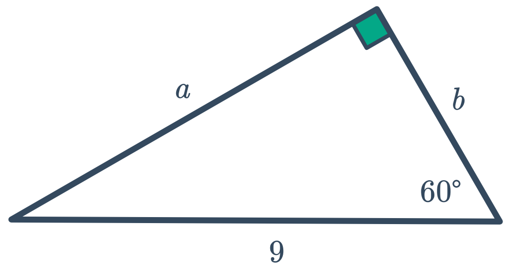
f



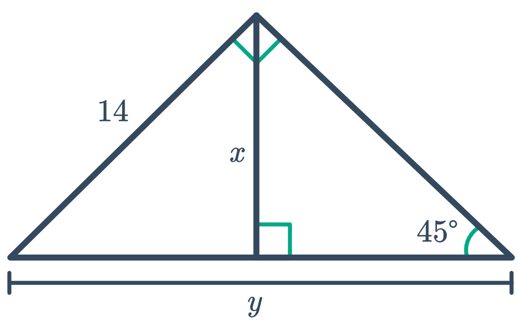
g



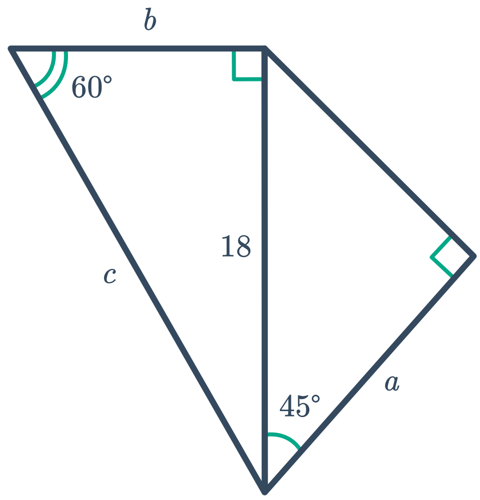
h



i

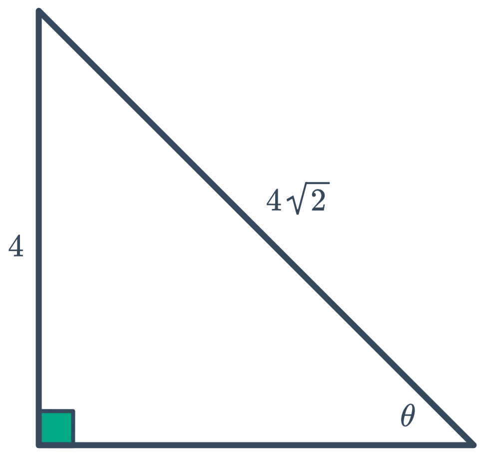


j

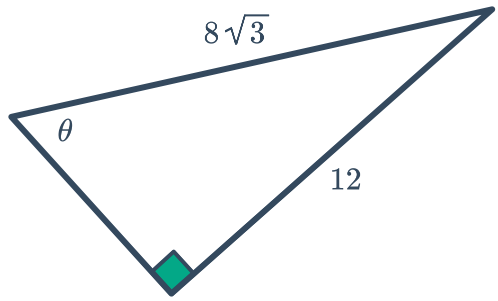


4 Use exact values to solve for θ in the follo[redacted]triangles:

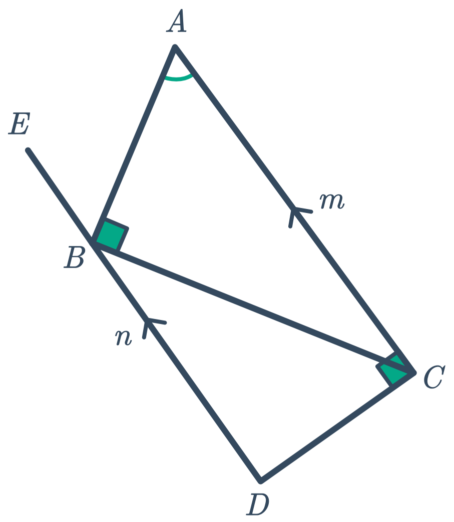
a



b



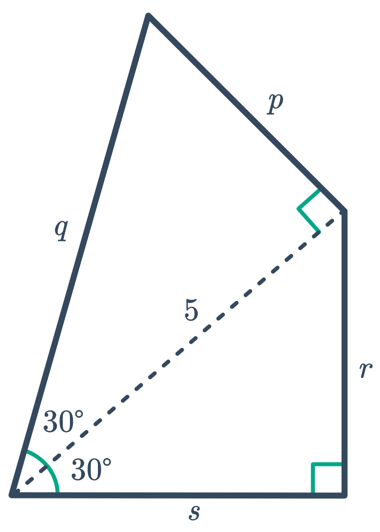
5 Consider the following diagram:



a If $\angle EBA$ is 60°, name two other angles that are also 60°.

b Given that CD has a length of $5\sqrt{7}$, calculate the exact values of m and n.

6 Consider the figure shown:

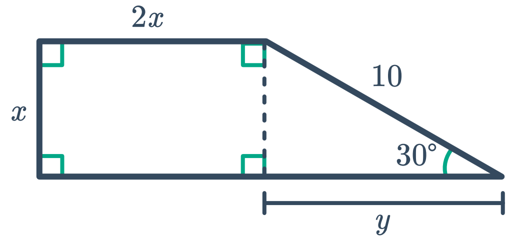


a Find the exact length of the following:

i p **ii** q **iii** r **iv** s

b Hence, find the exact perimeter of the quadrilateral.

7 Consider the given shape:



a Find x.

b Find the exact value of y.

c Hence, find the exact perimeter of the shape.